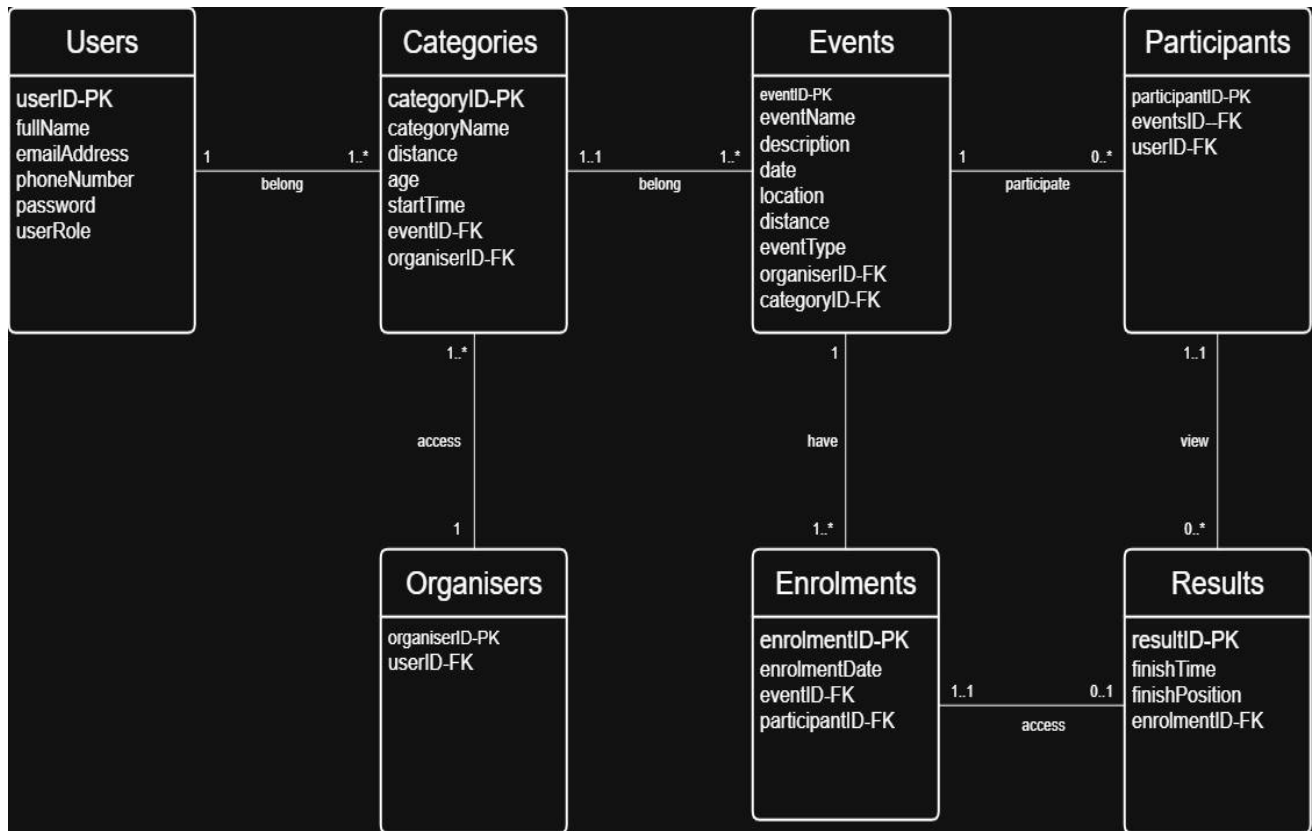




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PART1 Module Code: PROG6212
Submission Date: 04 September 2026

Section A

Entity Relationship Diagram (ERD)



Section B

API Endpoint Plan

HTTP method	Route	Description	Role required	Request body	Expected response
Post	/api/user/{ID}/register	Authenticate a new user to register as an organiser or participant.	None	{fullName, emailAddress, phoneNumber, password, userRole}	200 - User has successfully registered. 404 invalid user details are entered.
Post	/api/user/{ID}/login	Authenticate a user to login as an organiser or participant	None	{emailAddress, password}	200- User has logged in successfully. 404- Login details are invalid.

Get	/api/user/profile/view	Returns the user profile details. Use the view to see their own profile.	Any (logged in)	{ userID }	200- User's details have been retrieved.
Put	/api/user/profile/update	Updates the user's profile information.	Any (logged in)	{ fullName , password , userRole }	200- User profile has been updated. Conflict-session has expired.
Post	/api/events/create	Create a new event. Links with the logged-in organiser of the event.	Organiser	{ eventName, description, date, location, distance and eventType }	200 -Event created successfully . Conflict-event already exists.
Put	/api/events/update	Update an existing event.	Organiser	{ eventName, description, date, location, distance and eventType }	200 -Event updated successfully .
Delete	/api/events/delete	Delete an event	Organiser	{ eventID }	200- Event deleted successfully .
Get	/api/events/user/view	Returns events of a logged in organiser. Use the view to see all the events.	Organiser	{ eventID }	200 - Organiser has viewed an event.
Get	/api/events/user/view	Returns events of a logged in participant. Use the view to see all the events.	Participant	{ ParticipantID, eventID }	200 - Participant has viewed an event.

Get	/api/events/location	Shows all events	None	None	200- All events are successfully displayed.
Post	/api/events/{id}/categories	Create categories (age or distance) for each event.	Organiser	{eventId, category Name, categoryType }	200- Category successfully created for the event. 404-Event not found.
Get	/api/categories/view	Display all the available categories.	Any logged in	None	200-All event categories are displayed.
Post	/api/enrolment/create	Participant enters an event by selecting a category.	Participant	{eventId categoryID }	200- Category has been selected. Conflict- Participant already participating in an event.
Get	/api/enrolment/event/view	View the participant enrolment for an event.	Organiser	{ organiserID }	200- Enrolment for event is displayed. 404- Enrolment not found.
Post	/api/results/store	Capture finish times and finishing positions for participants.	Organiser	{enrolmentID, finishTime ,position, }	200-Finish times and positions has been entered successfully . Conflict invalid entered.
Get	/api/results/view	View my results for the event.	Participant	{resultID, finishTime , finishPosition, enrolment ID, participantID }	200- participant results have been successfully displayed. 404-Results

				}	are not found.
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Section C

SQL Database query

Create database Race_Day;
use Race_Day;

Create table Users(
userID int primary key,
fullName varchar(70) not null,
emailAddress varchar(70) not null,
phoneNumber varchar(25)not null,
password varchar(100) not null,
userRole varchar(20) not null
);

Create table Organisers(
organiserID int identity(1,1) primary key,
userID int,
Foreign key (userID) REFERENCES Users(userID)
);

Create table Event(
eventID int identity(1,1) primary key,
eventName varchar(60),
description varchar(70),
date date,
location varchar(70),
distance decimal(5,2),
eventType varchar(60),
organiserID int not null,

Foreign key (organiserID) REFERENCES Organisers (organiserID)
);

Create table Categories(
categoryID char(5) primary key,
categoryName varchar(50) not null,
distance decimal(5,2),
age int,
startTime time,
eventID int,
organiserID int,
Foreign key (eventID) REFERENCES Event(eventID),
Foreign key (organiserID) REFERENCES Organisers (organiserID)
);

Create table Participants(
participantID int identity(1,1) primary key,
eventID int,
userID int
Foreign key (eventID) REFERENCES Event (eventID),
Foreign key (userID) REFERENCES Users(userID)
);

Create table Enrolments(
enrolmentID int identity(1,1) primary key,
enrolmentDate date,
eventID int,
participantID int,
Foreign key (eventID) REFERENCES Event (eventID),
Foreign key (participantID) REFERENCES Participants (participantID),
);

Create table Results(

resultID int identity(1,1) primary key,

finishTime time,

finishPosition int,

enrolmentID int,

Foreign key (enrolmentID) REFERENCES Enrolments (enrolmentID)

);

Insert into Users

(userID,fullName,emailAddress,phoneNumber,password,userRole)

Values

(111,'Tshepang Mokoena','Tshe@gmail.com','0811165432','Tshe!12','Participant'),

(112,'Prince

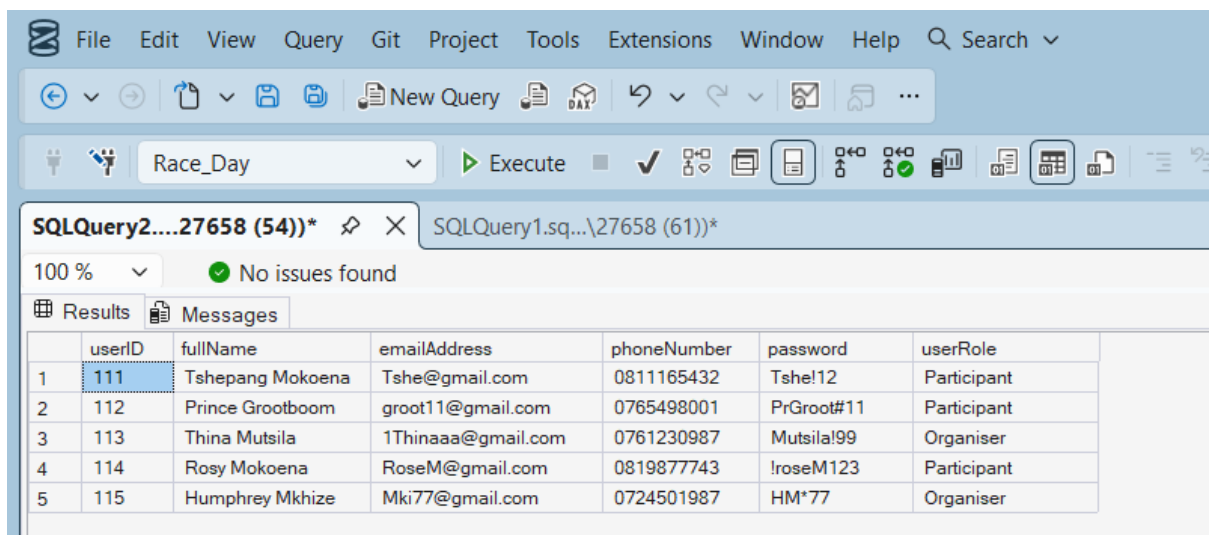
Grootboom','groot11@gmail.com','0765498001','PrGroot#11','Participant'),

(113,'Thina Mutsila','1Thinaaa@gmail.com','0761230987','Mutsila!99','Organiser'),

(114,'Rosy Mokoena','RoseM@gmail.com','0819877743','!roseM123','Participant'),

(115,'Humphrey Mkhize','Mki77@gmail.com','0724501987','HM*77','Organiser');

Select * From Users;



SQLQuery2....27658 (54)*

100 % No issues found

	userID	fullName	emailAddress	phoneNumber	password	userRole
1	111	Tshepang Mokoena	Tshe@gmail.com	0811165432	Tshe!12	Participant
2	112	Prince Grootboom	groot11@gmail.com	0765498001	PrGroot#11	Participant
3	113	Thina Mutsila	1Thinaaa@gmail.com	0761230987	Mutsila!99	Organiser
4	114	Rosy Mokoena	RoseM@gmail.com	0819877743	!roseM123	Participant
5	115	Humphrey Mkhize	Mki77@gmail.com	0724501987	HM*77	Organiser

Insert into Organisers

(userID)

Values

(111),

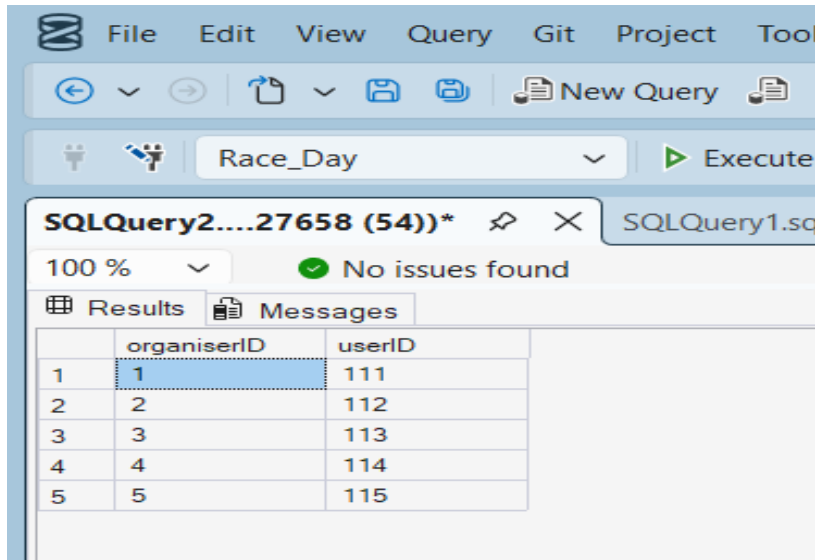
(112),

(113),

(114),

(115);

Select * From Organisers;



	organiserID	userID
1	1	111
2	2	112
3	3	113
4	4	114
5	5	115

Insert into Event

(eventName,description,date,location,distance,eventType,organiserID)

Values

('947 Ride Joburg','A premier 97km cycling race in Joburg streets','2026-08-11','Nasrec,Johannesburg',97.00,'Cycle',1),

('Cape town Cycle Walk','A 109 km cycling event that goes around Cape Town','2026-04-21','Cape Town City Centre',109.00,'Cycle',2),

('Soweto Marathon', 'A annual long-distance running event throgh historic streets of Soweto','2026-09-07','Nasrec Expo Centre,Johannesburg',42.2,'Run',3),

('702 Walk the Talk','Participation event in Johannesburg to walk and celebrate city streets','2026-11-26','Doornfontein,Johannesburg',5.00,'Walk',4),

('The Color Run','A five km fun run with vibrant participants','2026-08-29','Sowto Theatre Forecourt',5.00,'Run',5);

Select * From Event;

	eventID	eventName	description	date	location	distance	eventType	organiserID
1	1	947 Ride Joburg	A premier 97km cycling race in Joburg streets	2026-08-11	Nasrec,Johannesburg	97.00	Cycle	1
2	2	Cape town Cycle Walk	A 109 km cycling event that goes around Cape Town	2026-04-21	Cape Town City Centre	109.00	Cycle	2
3	3	Soweto Marathon	A annual long-distance running event through historic streets of Soweto	2026-09-07	Nasrec Expo Centre,Johannesburg	42.20	Run	3
4	4	702 Walk the Talk	Participation event in Johannesburg to walk and celebrate city streets	2026-11-26	Doornfontein,Johannesburg	5.00	Walk	4
5	5	The Color Run	A five km run fun with vibrant participants	2026-08-29	Sowto Theatre Forecourt	5.00	Run	5

Insert into Categories

(categoryID,categoryName,distance,age,startTime,eventID,organiserID)

Values

('C001','Full marathon',97.00,20,'06:00',1,1),

('C002','109 km Cycle',109.00,13,'07:00',2,2),

('C003','42.2 km Run',42.2,20,'07:00',3,3),

('C004','Under 20',5.00,9,'06:30',4,4),

('C005','5km Run',5.00,4,'08:00',5,5);

Select * From Categories;

	categoryID	categoryName	distance	age	startTime	eventID	organiserID
1	C001	Full marathon	97.00	20	06:00:00.0000000	1	1
2	C002	109 km Cycle	109.00	13	07:00:00.0000000	2	2
3	C003	42.2 km Run	42.20	20	07:00:00.0000000	3	3
4	C004	Under 20	5.00	9	06:30:00.0000000	4	4
5	C005	5km Run	5.00	4	08:00:00.0000000	5	5

Insert into Participants

(eventID,userID)

Values

(1,111),

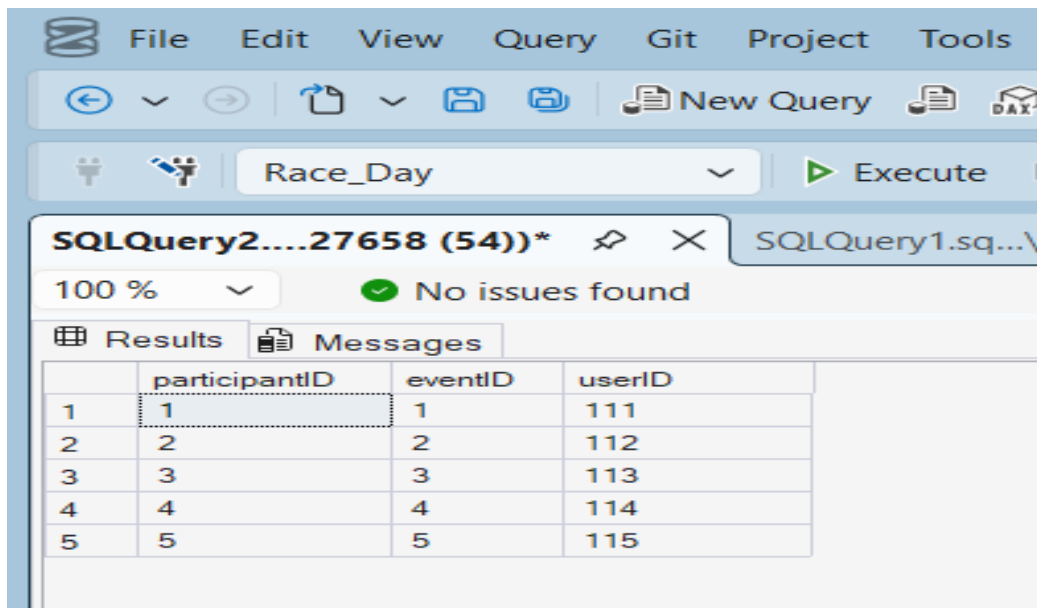
(2,112),

(3,113),

(4,114),

(5,115);

Select * From Participants;



The screenshot shows the SQL Server Enterprise Manager interface. The top menu bar includes File, Edit, View, Query, Git, Project, and Tools. Below the menu is a toolbar with navigation and action icons. The main pane displays a query window titled 'SQLQuery2....27658 (54))*' with a 'Race_Day' dropdown and an 'Execute' button. The query results are shown in a table with the following data:

	participantID	eventID	userID
1	1	1	111
2	2	2	112
3	3	3	113
4	4	4	114
5	5	5	115

Insert into Enrolments

(enrolmentDate,eventID,participantID)

Values

('2026-04-21',2,2),

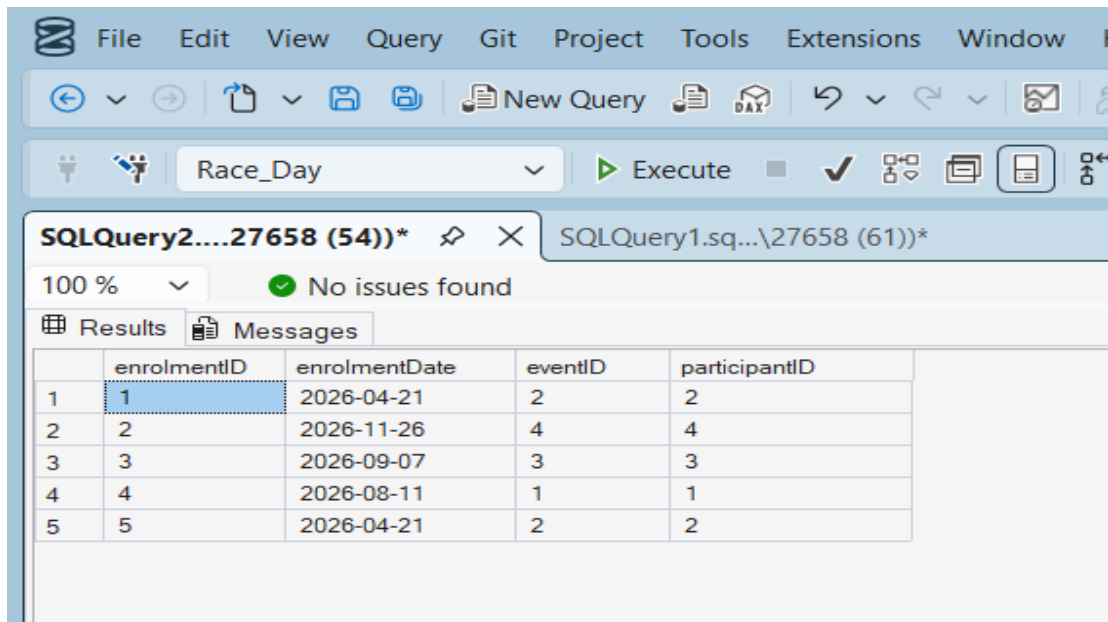
('2026-11-26',4,4),

('2026-09-07',3,3),

('2026-08-11',1,1),

('2026-04-21',2,2);

Select * From Enrolments;



SQLQuery2....27658 (54))*

100 % No issues found

Results Messages

	enrolmentID	enrolmentDate	eventID	participantID
1	1	2026-04-21	2	2
2	2	2026-11-26	4	4
3	3	2026-09-07	3	3
4	4	2026-08-11	1	1
5	5	2026-04-21	2	2

Insert into Results

(finishTime,finishPosition,enrolmentID)

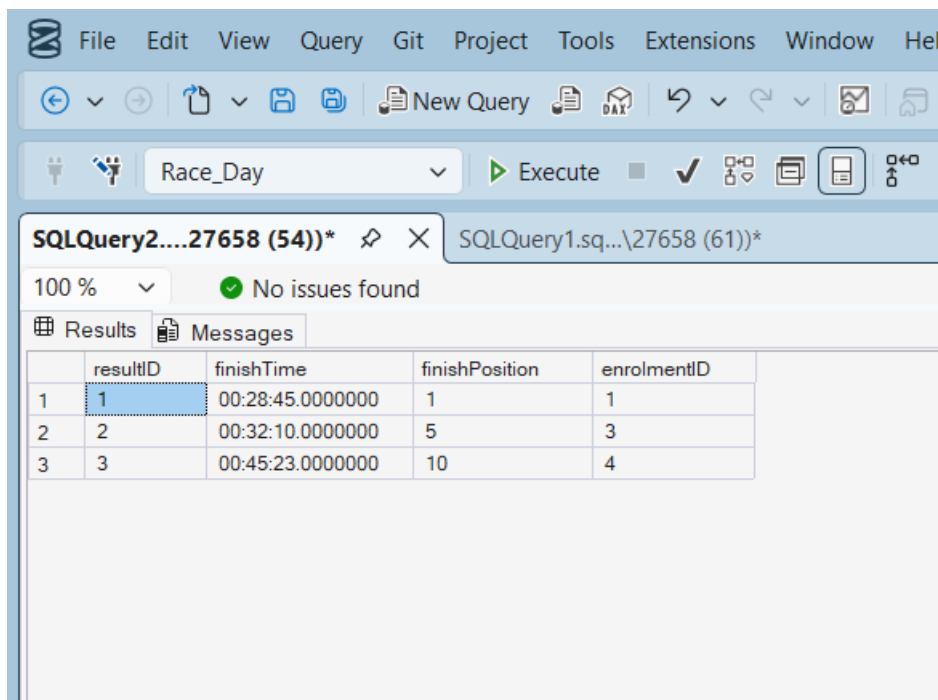
Values

('00:28:45',01,1),

('00:32:10',05,3),

('00:45:23',10,4);

Select * From Results;



SQLQuery2....27658 (54))*

100 % No issues found

Results Messages

	resultID	finishTime	finishPosition	enrolmentID
1	1	00:28:45.0000000	1	1
2	2	00:32:10.0000000	5	3
3	3	00:45:23.0000000	10	4

YouTube Video Link: https://youtu.be/_VjFfUsYEt4

Proof of YouTube video:

The screenshot shows a YouTube video player with the video title "PART1". The video content is a Microsoft Word document titled "PROG6212_POE.docx". The document displays an Entity Relationship Diagram (ERD) for a system. The ERD is divided into two sections: Section A and Section B.

Section A: Entity Relationship Diagram (ERD)

Entities and their attributes:

- Users**: userID-PK, surname, emailAddress, phoneNumber, password, userID
- Categories**: categoryID-PK, categoryName, distance, age, eventTime, eventID-FK, organiserID-FK
- Events**: eventID-PK, eventName, description, date, location, distance, eventTime, eventID-FK, categoryID-FK
- Participants**: participantID-PK, eventID-FK, userID-FK
- Organisers**: organiserID-PK, userID-FK
- Enrolments**: enrolmentID-PK, enrolmentDate, eventID-FK, participantID-FK
- Results**: resultID-PK, resultTime, resultPosition, enrolmentID-FK

Relationships:

- Users** (1) **Categories** (1,1) - relationship: belong
- Categories** (1,1) **Events** (1,1) - relationship: belong
- Events** (1) **Participants** (1,1) - relationship: participate
- Organisers** (1) **Events** (1,1) - relationship: organise
- Enrolments** (1,1) **Results** (1,1) - relationship: enrol
- Participants** (1,1) **Enrolments** (1,1) - relationship: participate

Section B

The video player interface includes a search bar, a video title "PART1", a channel name "Wanga Mutsila" with 1 subscriber, and buttons for "Analytics", "Edit video", "Share", "Promote", and "Download".